RESEARCH LETTER

Channel attention in lightweight hyperspectral CNNs: an equivalence-tested comparison

Swati Hira[a], Devguru Tiwari[a], Pratibha Kokardekar[b] and S. R. Sathe[c]

[a]Department of Computer Science and Engineering, Indian Institute of Information Technology Nagpur, Nagpur, India; [b]Ramdeobaba University, Nagpur, India; [c]Visvesvaraya National Institute of Technology, Nagpur, India

ARTICLE HISTORY
Compiled October 24, 2026

ABSTRACT
Channel-attention blocks are often added to lightweight networks for hyperspectral image classification, with gains reported on random pixel splits. We test whether such gains are real and whether they transfer. Seven channel-modulation blocks (SE, ECA, CBAM, GCT, SRM and two variants of SNSM, a dispersion-aware probe) are compared on one backbone across five scenes, with up to fifteen paired seeds, Holm-corrected Wilcoxon tests and equivalence tests with a margin of ± 1 point of average accuracy (AA). On random 5% splits, blocks differ only on Indian Pines, where dispersion-based gates exceed the others by 1.8–3.8 AA points through two single-pixel classes. This gap vanishes once every class has 15 training pixels, does not reappear under the same scarcity on other scenes, shrinks at a higher learning rate, and is matched by a deeper plain CNN. On random splits of WHU-Hi HanChuan all blocks are equivalent. On spatially disjoint splits accuracy falls by 16–45 points, no block is consistently best, and on Pavia University four blocks, including the dispersion-based ones, are significantly worse than the plain CNN. Class counts, host network, training setting and evaluation protocol shape reported accuracy more than the attention block.

KEYWORDS
Hyperspectral image classification; channel attention; equivalence testing; spatial leakage; few labels

1. Introduction

Hyperspectral images record hundreds of contiguous bands. This makes pixel-level land-cover classification possible, but when labelled samples are few it exposes classifiers to the Hughes phenomenon (Hughes 1968; Bioucas-Dias et al. 2013). Convolutional neural networks (CNNs) that operate on spatial–spectral patches are the usual lightweight choice (Li et al. 2019; Audebert, Le Saux, and Lefevre 2019; Makantasis et al. 2015; Chen et al. 2016; Zhong et al. 2018; Roy et al. 2020), and transformers add global context at a higher cost (Hong et al. 2022; Sun et al. 2022). Channel attention is a cheap way to make such networks input-dependent: squeeze-and-excitation (SE) predicts one scale per channel from pooled means through fully connected (FC) layers (Hu, Shen, and Sun 2018); ECA replaces the FC layers by a one-dimensional

CONTACT Swati Hira. Email: shira@iiitn.ac.in

convolution (Wang et al. 2020); CBAM adds a spatial gate (Woo et al. 2018); GCT normalises channel statistics and gates them with $1 + \tanh(\cdot)$ (Yang et al. 2020); and SRM pools channel means and standard deviations (Lee, Kim, and Nam 2019). Hyperspectral networks have adopted these blocks widely (Mei et al. 2019; Haut et al. 2019; Li et al. 2020; Zhu et al. 2021; Roy et al. 2021).

Most reported gains come from random pixel splits. With patch-based models, training and test patches then overlap, so test accuracy partly measures memorised spatial context (Liang et al. 2017; Nalepa, Myller, and Kawulok 2019). This leakage has been surveyed (Feng et al. 2023), disjoint sampling changes the accuracy of deep models (Ahmad et al. 2024), and spatially separated splits lower the overall accuracy of CNNs and transformers on Pavia University by about 40 points (Alkhatib 2026). These studies compare whole architectures. They do not test whether the *choice of channel-attention block* matters, and a block that "brings no significant gain" is usually reported without showing that its effect is actually small. In computer vision, careful re-evaluations have found that training and scaling choices can outweigh architectural changes (Bello et al. 2021), that many reported improvements shrink under a fair protocol (Musgrave, Belongie, and Lim 2020), and that seed-to-seed variance must be accounted for before ranking methods (Bouthillier et al. 2021); we apply this kind of check to channel attention in hyperspectral classification.

Our contributions are:

- a paired, equivalence-tested comparison of seven blocks on one backbone across five scenes and eight protocols (Sections 2–3);
- evidence that the one large difference, on random Indian Pines splits, is an effect of single-pixel classes, host network and learning rate rather than of the block (Section 4);
- a three-step check for evaluating new blocks.

2. Blocks compared

All blocks are inserted in the same backbone. A 3×3 convolution maps the B input features to $C = 64$ channels (batch normalisation (Ioffe and Szegedy 2015), ReLU). Two residual blocks (He et al. 2016) follow, each computing $\mathbf{h} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{x})))$ and $\mathbf{y} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathcal{M}(\mathbf{h}))) + \mathbf{x})$, where $\mathcal{M}$ is the block under study ($\mathcal{M}$ = identity for the plain CNN). Global average pooling, dropout (0.4) (Srivastava et al. 2014) and a linear layer produce the class scores. SE, ECA (kernel 3, the adaptive value for 64 channels) and CBAM (reduction 16, 7×7 spatial gate) output a sigmoid scale per channel; GCT uses its ℓ_2 embedding and an identity start; SRM gates each channel from its mean and standard deviation through a channel-wise FC layer and batch normalisation.

SNSM as a probe. To test which channel statistic matters, we also include spectral-neighbourhood self-modulation (SNSM), a block whose inputs can be switched on and off. For every channel c it computes the spatial mean, the spatial variance and the difference of the neighbouring channels' means, standardises each across channels, and filters the three with two one-dimensional convolutions along the channel axis (8 hidden channels, kernel 3, dilation 2) into a scale $\gamma_c = 1 + \frac{1}{2}\tanh(z_{1,c})$ and a shift $\beta_c = z_{2,c}$, so that $\mathcal{M}(\mathbf{h})_c = \gamma_c \mathbf{h}_c + \beta_c$. The second convolution starts at zero, so the block is initially the identity; it adds 130 parameters per block. SNSM (γ) drops the shift. The block diagram and the code are in the supplement and repository.

Table 1. Benchmark scenes and the random 5% splits (seed 42; split sizes are the same for every seed).

Scene	Sensor	Size (pixels)	Bands	GSD (m)	Classes	Labelled	Train/test (random)
Indian Pines (IP)	AVIRIS	145×145	200	20	16	10,249	505/9,744
Pavia University (PU)	ROSIS	610×340	103	1.3	9	42,776	2,135/40,641
Kennedy Space Center (KSC)	AVIRIS	512×614	176	18	13	5,211	256/4,955
Botswana (BW)	Hyperion	1476×256	145	30	14	3,248	156/3,092
WHU-Hi HanChuan (WHU)	Headwall Nano-Hyperspec	1217×303	274	0.109	16	257,530	12,868/50,000

3. Data and protocols

Data. We use Indian Pines (IP), Pavia University (PU), Kennedy Space Center (KSC), Botswana (BW) and the larger, more recent WHU-Hi HanChuan (WHU) (Baumgardner, Biehl, and Landgrebe 2015; Gamba 2004; Grupo de Inteligencia Computacional 2020; Zhong et al. 2020) (Table 1). Each band is smoothed with a 5×5 Wiener filter, principal component analysis (PCA) (Jolliffe and Cadima 2016), fitted without labels on all pixels, keeps 30 components, and the db4 wavelet approximation of each component (Mallat 1989) is resized and appended, giving $B = 60$ features. Each labelled pixel is represented by its $15 \times 15 \times B$ patch. *Note on KSC:* the distributed file stores small negative values (-27 to -1) as unsigned 16-bit integers, which read as 65,509–65,535 and dominate the PCA. Kakde et al. (2018) treated them as saturation and set them to zero, while loaders that read the file as unsigned and rescale it globally, such as that of DeepHyperX (Audebert, Le Saux, and Lefevre 2019), keep the wrapped values. We read the file as signed integers; read as unsigned, seven of our CNN variants reach only 77–92% AA on KSC instead of 99.2–99.5%.

Protocols. *Random 5%*: for each class, 5% of the labelled pixels (at least one) train and the rest test, giving 505/9,744 (IP), 2,135/40,641 (PU), 256/4,955 (KSC), 156/3,092 (BW) and 12,868/50,000 (WHU, test set capped at 50,000 pixels) pixels; Alfalfa, Grass-pasture-mowed and Oats then have two, one and one training pixels. *Smallest 15* (IP): the same split with only those three classes raised to 15 training pixels. *Random 50/15* (IP): a random split with the class counts of the standard split, 15 training pixels for those three classes and 50 for each other class (695/9,554). *Standard* (IP): the fixed training and test sets distributed with SpectralFormer (Hong et al. 2022) (695/9,671), whose training pixels come from separate regions but whose test patches still overlap training patches at 15×15. *Smallest 1/1/2* (PU, KSC): the random 5% split with the three smallest classes of each scene reduced to 1, 1 and 2 training pixels, as in IP. *Hardest 1/1/2* (PU, KSC): the same, but for the three classes on which the plain CNN is least accurate under random 5% splits. *Patch-disjoint* (PU, KSC, WHU): for each class, the 5% of its pixels nearest a random centre form a compact training cluster, and every test pixel within 14 pixels of any training pixel is discarded, so no test patch overlaps a training patch (Figure 1); 33,847–36,453 (PU) and 3,077–3,513 (KSC) test pixels remain, and the WHU test set is again capped at 50,000 pixels. This rule leaves no test pixels for five IP classes, and the standard PU and Houston splits were no longer available, so these three scenes carry the patch-disjoint evaluation. *Block-disjoint* (PU, KSC): a milder spatial split in which the scene is cut into 16×16 blocks visited in one random order per seed and each class takes its 5% training pixels block by block, so that training pixels are scattered over many small blocks rather than one cluster; the same 14-pixel buffer removes overlapping test patches. BW was evaluated on random splits only. *Robustness checks*: a deeper, wider host with four residual blocks of 128 channels (IP random 5%, KSC patch-disjoint); learning rates of 3×10^{-4} and 3×10^{-3} (IP random 5%); the blocks applied to the raw bands in wavelength order, without PCA (IP, PU, KSC random 5%); and PCA fitted



Figure 1. Patch-disjoint split of Pavia University (seed 42). Training pixels form one compact cluster per class; test pixels whose 15×15 patch would overlap a training patch are excluded.

on the training pixels only (KSC patch-disjoint).

Training and statistics. Every model sees the same features and splits for a given seed. Training uses cross-entropy, AdamW (Loshchilov and Hutter 2019) (learning rate 10^{-3}, weight decay 10^{-4}), step decay (factor 0.5 every 40 epochs), batch size 64 and exactly 100 epochs, without augmentation or early stopping; the final model is evaluated once. All eight backbone variants use 15 seeds (42–56), except on the random 5% splits of PU, KSC and BW (five seeds), on WHU and with the deeper host (ten seeds), with the learning rate 3×10^{-4} and on raw bands (five seeds); on the standard split the seed changes only the initialisation and batch order. For each of the 19 settings we compare SNSM (γ+β) with the seven other blocks by paired same-seed differences of AA; the supplement repeats every comparison with the plain CNN as the reference, so that no block is privileged. Differences are tested with two-sided Wilcoxon signed-rank tests, Holm-corrected within each setting. With five seeds the smallest attainable Wilcoxon p is 0.0625, so those settings are descriptive only. The study is exploratory: several protocols were added to test explanations suggested by earlier results, which is why we also report a correction across all settings. Across all 133 comparisons, Benjamini–Hochberg control of the false discovery rate at 5% gives 18 discoveries, against 22 with the per-setting Holm correction. Equivalence is tested with paired two one-sided t-tests (TOST, $\alpha = 0.05$), because the Wilcoxon test gives no interval: a difference is equivalent if its 90% t interval lies within ± 1 AA point. We chose this margin as less than half the median seed-to-seed standard deviation of AA on random IP splits (2.2 points), a difference no single run could resolve; the supplement repeats the equivalence tests at ± 0.5 and ± 2 points. Runs used PyTorch (Paszke et al. 2019) on NVIDIA Tesla T4 GPUs, except ten KSC patch-disjoint seeds,

Table 2. Random 5% splits: OA and AA (%, mean ± std) of the eight backbone variants (15 seeds on IP, five elsewhere). Best mean per column in bold. External baselines (HybridSN, SSFTT) and an FC (γ+β) gate are in the supplement.

Method	IP		KSC		PU		BW	
	OA	AA	OA	AA	OA	AA	OA	AA
Plain CNN	96.28±0.79	89.49±2.44	99.65±0.34	99.51±0.36	**99.59±0.07**	99.01±0.16	99.09±0.55	98.87±0.90
SE	**96.34±0.62**	90.04±2.18	99.61±0.28	99.43±0.38	99.53±0.19	98.80±0.56	98.95±0.82	98.76±1.07
ECA	96.28±0.68	90.87±2.53	99.61±0.47	99.40±0.66	99.58±0.10	99.00±0.20	**99.13±0.76**	**98.90±1.08**
CBAM	96.19±0.60	91.58±1.49	99.54±0.56	99.23±0.83	99.58±0.06	**99.11±0.18**	98.99±0.87	98.78±1.20
GCT	96.28±0.80	90.07±2.34	**99.69±0.26**	**99.55±0.28**	99.58±0.08	98.99±0.18	99.11±0.56	98.88±0.91
SRM	96.20±0.71	93.04±2.14	99.69±0.36	99.51±0.49	99.55±0.07	98.92±0.17	98.87±0.63	98.67±0.97
SNSM (γ)	96.17±0.73	92.89±2.00	99.69±0.26	99.50±0.39	99.49±0.05	98.91±0.16	99.10±0.62	98.89±0.96
SNSM (γ+β)	96.27±0.67	**93.33±1.68**	99.63±0.55	99.45±0.73	99.47±0.05	98.93±0.16	99.04±0.69	98.85±1.03

Table 3. Indian Pines: AA (%, mean ± std over 15 seeds) of the eight backbone variants under four protocols. The last row is the range of the block means.

Block	Random 5%	Random, smallest 15	Random, 50/15	Standard split
Plain CNN	89.49±2.44	96.03±0.76	97.83±0.52	97.24±0.23
SE	90.04±2.18	95.93±0.64	**97.90±0.44**	97.25±0.26
ECA	90.87±2.53	96.10±0.76	97.79±0.49	97.15±0.22
CBAM	91.58±1.49	95.83±0.67	97.78±0.51	97.45±0.29
GCT	90.07±2.34	96.07±0.75	97.82±0.53	97.23±0.22
SRM	93.04±2.14	96.01±0.81	97.76±0.57	**97.50±0.22**
SNSM (γ)	92.89±2.00	96.13±0.72	97.77±0.54	97.37±0.22
SNSM (γ+β)	**93.33±1.68**	**96.43±0.66**	97.82±0.43	97.48±0.26
Range of block means	3.84	0.60	0.14	0.35

which ran on an RTX 3050 laptop GPU for all blocks and are pooled with the five T4 seeds; each paired difference compares blocks run on the same GPU.

4. Results

Random 5% splits. On PU, KSC and BW the eight variants lie within 0.32 AA points of each other (Table 2), and on WHU, the largest scene, within 0.06 points, with every comparison equivalent within ±1 point (Tables 5 and 4). On these scenes AA is close to 99%, and WHU has 12,868 training pixels, so this equivalence partly reflects a ceiling and an ample training set; the few-label case is tested on IP and by the scarcity protocols below. Only IP separates them: SNSM (γ+β) is significantly better than the plain CNN, SE, ECA, CBAM and GCT, by 1.75–3.84 points, but not than SRM or SNSM (γ) (Table 4), while the overall accuracy of all eight differs by only 0.17 points. The gap lies almost entirely in the two classes with a single training pixel and 19 and 27 test pixels: SRM and the SNSM variants reach 80.0–84.4% on Grass-pasture-mowed and 66.3–77.2% on Oats, against 62.7–78.5% and 36.8–56.1% for the other blocks, whereas on Alfalfa (two training pixels) all blocks lie within 3.3 points and their mean over the other 13 classes differs by less than one point. With so few test pixels, one or two pixels move these class accuracies by about 4–11 points (1/27 to 2/19), which is why a small overall difference becomes a large difference in AA.

Class counts. Raising only the three smallest IP classes to 15 training pixels shrinks the spread of block means from 3.84 to 0.60 points (Table 3). SNSM (γ+β) keeps a small edge of 0.30–0.60 points over every block, significant per setting but for only 4 of the seven blocks under the global correction, and every one of these differences is also equivalent within ±1 point (Table 4). Being both significant and equivalent, the edge

Table 4. Paired AA differences SNSM (γ+β) − block (points). *: Holm-corrected Wilcoxon $p < 0.05$ within the row; ‡: also a discovery under Benjamini–Hochberg across all 133 comparisons; †: equivalent within ±1 point (paired TOST, $\alpha = 0.05$). Rows with five seeds cannot reach $p < 0.05$. 95% intervals, OA differences, other margins and comparisons with the plain CNN are in the supplement.

Setting	Plain	SE	ECA	CBAM	GCT	SRM	SNSM (γ)	Seeds
IP, random 5%	+3.84*‡	+3.29*‡	+2.45*‡	+1.75*‡	+3.26*‡	+0.29	+0.44	15
IP, smallest 15	+0.39*‡†	+0.50*‡†	+0.33*†	+0.60*‡†	+0.35*†	+0.41*†	+0.30*‡†	15
IP, random 50/15	-0.01†	-0.08†	+0.04†	+0.05†	+0.01†	+0.07†	+0.06†	15
IP, standard	+0.24†	+0.23†	+0.33*‡†	+0.04†	+0.25†	-0.02†	+0.12†	15
PU, smallest 1/1/2	-0.59	-0.74	-1.19*‡	-0.30†	-1.14	-0.55	-1.59*‡	15
KSC, smallest 1/1/2	-0.24†	+0.79	+0.39	+0.54	-0.21†	+1.41	+0.03	15
PU, hardest 1/1/2	+0.97	+0.98	+0.64	+0.96	+0.77	+0.79	-0.82	15
KSC, hardest 1/1/2	+0.71	+0.88	+0.20†	+1.14	+0.77	+0.66	+0.90	15
WHU, random 5%	-0.06†	-0.01†	-0.02†	+0.00†	+0.00†	-0.06†	-0.03†	10
PU, patch-disjoint	+0.31	-0.47	-0.39	+1.03	+0.92	+1.33	+0.03	15
KSC, patch-disjoint	+3.46*‡	-0.19	+1.86	+0.91	+3.51*‡	+4.25*	+1.40	15
WHU, patch-disjoint	-2.01	-2.39	-1.77	-1.85	-1.32	-0.84	-0.46	10
PU, block-disjoint	-3.77*‡	-2.24	-2.95*‡	+0.72	-4.04*‡	-1.14	-1.16	15
KSC, block-disjoint	-0.63	-1.66	-0.99	-0.56	-0.56	+1.64	-1.24	15
IP, random 5%, deeper host	-0.53	+0.14†	+0.18†	-0.35	-0.51	-0.87	-0.19	10
KSC, patch-disjoint, deeper host	-1.88	-0.80	-2.87	-1.35	-1.63	+0.34	-2.00	10
KSC, patch-disjoint, train-only PCA	+0.33	-0.98	-0.01	-2.38*‡	+0.54	+1.65	-0.38	15
IP, random 5%, lr 3×10^{-4}	+8.16	+7.42	+8.26	+6.12	+8.06	+5.37	+5.27	5
IP, random 5%, lr 3×10^{-3}	+0.98	+1.48	+0.56	+0.80	+0.83	+0.28	-0.03	15

Table 5. PU, KSC and WHU: AA under random 5% splits and OA and AA under patch-disjoint splits (%, mean ± std; PU and KSC: five random and 15 patch-disjoint seeds; WHU: ten seeds each). The last row is the range of the block means.

Block	PU			KSC			WHU		
	AA, random	OA, disjoint	AA, disjoint	AA, random	OA, disjoint	AA, disjoint	AA, random	OA, disjoint	AA, disjoint
Plain CNN	99.01±0.16	55.31±9.82	67.03±5.03	99.51±0.36	75.13±4.97	63.55±6.41	99.26±0.17	57.57±4.48	55.30±4.38
SE	98.80±0.56	55.91±9.73	**67.81±4.39**	99.43±0.38	76.79±4.44	**67.20±6.66**	99.21±0.24	**57.92±6.73**	**55.67±5.99**
ECA	99.00±0.20	56.30±9.17	67.73±4.43	99.40±0.66	76.40±4.81	65.15±5.86	99.23±0.21	56.29±5.44	55.06±4.84
CBAM	**99.11±0.18**	56.11±10.31	66.31±4.80	99.23±0.83	76.68±4.74	66.10±7.39	99.21±0.19	56.28±5.80	55.14±3.81
GCT	98.99±0.18	54.59±9.68	66.42±4.57	**99.55±0.28**	75.20±5.16	63.49±6.80	99.20±0.22	55.30±5.50	54.61±5.12
SRM	98.92±0.17	56.02±9.99	66.00±4.73	99.51±0.49	74.36±4.22	62.75±6.38	**99.27±0.22**	57.67±7.09	54.13±5.37
SNSM (γ)	98.91±0.16	55.41±9.44	67.31±4.57	99.50±0.39	76.86±4.19	65.60±7.15	99.23±0.19	55.00±4.31	53.75±4.63
SNSM (γ+β)	98.93±0.16	**56.80±11.83**	67.34±4.95	99.45±0.73	**77.11±3.54**	67.01±6.00	99.21±0.19	55.06±3.91	53.29±5.08
Range of block means	0.31	2.21	1.80	0.32	2.75	4.44	0.06	2.92	2.39

is real but smaller than the margin; within ±0.5 point only 1 of the seven comparisons is equivalent (supplement). With 15–50 training pixels per class, in the random 50/15 split or the standard split, the block means span 0.14 and 0.35 points and all fourteen comparisons are equivalent within ±1 point; only the difference against ECA on the standard split (0.33 points) is also significant. Because the random 50/15 split keeps the random geometry and changes only the class counts, the IP gap is an effect of the single-pixel classes, not of spatial leakage.

Scarcity elsewhere. Reducing the three smallest classes of PU and KSC to 1, 1 and 2 training pixels lowers AA by about 15 and 7 points, but the dispersion-based gates do not gain (Table 4): on PU, SNSM (γ+β) has the lowest mean AA and is significantly worse than ECA and SNSM (γ); on KSC no difference is significant. Making the three *hardest* classes scarce instead gives the same picture: all differences stay within 1.1 points and none is significant. Per-block AA is in the supplement.

Patch-disjoint splits. Removing patch overlap lowers AA by 32.0 points on PU, 34.3 on KSC and 44.6 on WHU, averaged over blocks, and OA to 55–57%, 74–77% and 55–58% (Table 5). Block means span 1.80 (PU), 4.44 (KSC) and 2.39 (WHU) points, but the seed-to-seed standard deviation is 3.8–7.4 points and the median 95% interval of a paired difference is 4.0 (PU) and 3.9 (KSC) points wide, so differences of about two points can be neither confirmed nor excluded: no comparison is equivalent within ±1 point, and at most one per scene within ±2 points. On PU no difference is significant; on KSC, SNSM (γ+β) is significantly better than the plain CNN, GCT and

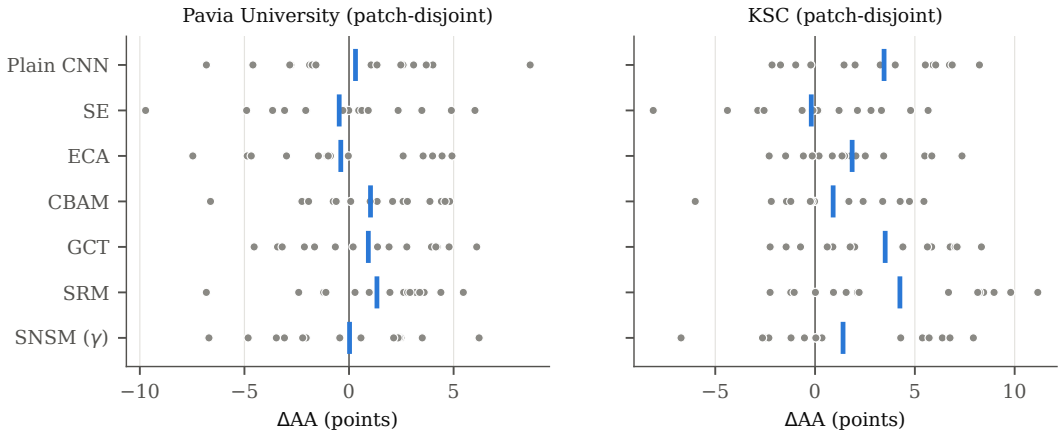


Figure 2. Paired same-seed AA differences SNSM (γ+β) − block on the patch-disjoint splits of PU and KSC. Grey dots are the 15 seeds; the blue bar is their mean. WHU (ten seeds) is in Table 4.

SRM (SRM not under the global correction), but not than SE, ECA, CBAM or SNSM (γ), and SE has the highest mean AA (Figure 2); against the plain CNN, only SE and SNSM (γ+β) are significantly better on KSC; on WHU, SNSM (γ+β) has the lowest mean AA (53.3–55.7% across blocks) and no difference is significant. The ranking also changes between protocols: SRM, tied best on random 5% IP, is the lowest on patch-disjoint KSC. The milder block-disjoint split lowers AA by 18 (PU) and 28 (KSC) points instead, to 78.7–83.5% and 69.2–72.5%. On KSC no difference is significant; on PU the plain CNN and GCT are the best blocks, SNSM (γ+β) is 3.8 points below the plain CNN, and CBAM, SRM and both SNSM variants are significantly worse than the plain CNN (supplement). GCT stays close to the plain CNN throughout because its trained gate departs from the identity by at most 0.07 on average.

Host network, learning rate and features. In the deeper, wider host the Indian Pines block means span only 1.05 points, SNSM (γ+β) is no longer the best block, the plain CNN reaches 93.8% AA, matching the best block in the shallow host (93.3%; Welch t-test $p = 0.46$, as the seeds differ), and on patch-disjoint KSC no difference is significant (Table 4; per-block AA in the supplement). Fitting PCA on the training pixels only leaves patch-disjoint KSC accuracy unchanged (62.9–67.0% against 62.8–67.2% AA across blocks), so the scene-level PCA does not inflate these results. The learning rate changes the size of the Indian Pines gap. With 3×10^{-3} the plain CNN rises from 89.5% to 92.2% AA, SNSM (γ+β) exceeds the other blocks by at most 1.5 points, still on 6 of seven comparisons, but no difference is significant (15 seeds). With 3×10^{-4} all blocks undertrain in 100 epochs and SNSM (γ+β) degrades least, by 5.3–8.3 points over the others (five seeds, descriptive only). Giving each block its best of the three learning rates, the block means span 91.7–93.3%, SRM and the SNSM variants remain the top three, and SNSM (γ+β) leads the plain CNN by 1.2 points. The single-pixel-class gap is thus partly a property of the host network and the optimisation setting.

Which statistic matters. Removing inputs from the SNSM probe on random 5% IP (five seeds, descriptive only; supplement) shows that the variance carries the gap: with the mean alone AA falls by 1.16 points, adding the gradient to the mean does not recover it, and mean and variance alone are 0.39 points better than the full probe (4 of five seeds), so the gradient input is not needed; without the zero initialisation AA is lower on all five seeds. This matches the tie between SNSM and SRM, the only other

block that gates on dispersion. Applied to the raw bands in wavelength order instead of the PCA features (five seeds; supplement), the blocks show the same pattern: within 0.7 AA points of each other on PU and KSC, while on IP SNSM (γ+β) exceeds the others by 3.9–4.9 points. All blocks add at most a few thousand parameters; SNSM's many small operations make its batch-1 latency about 3.7 times that of the plain CNN (supplement).

5. Discussion and limitations

Across five scenes and eight protocols, the choice of channel-attention block produced a large and consistent difference in one setting only: random 5% splits of Indian Pines, where two classes have a single training pixel. There, gates that see channel dispersion recognised those classes more often, but the size of the gap depended on the host network and the learning rate. With 15 or more training pixels per class, and on random splits of the large WHU-Hi scene, all blocks were equivalent within ± 1 AA point, and making the smallest or the hardest classes of Pavia University and KSC equally scarce did not favour dispersion-based gates. Spatial leakage moved accuracy far more than any block did, and without it the dispersion-based gates had no consistent advantage and on block-disjoint PU were worse than no attention at all: without patch overlap every block lost 16–45 AA points, the ranking of the blocks changed, and the remaining differences were mostly not significant, although differences of about two points could not be excluded. A gain reported for a new block on a random split should therefore be checked for its dependence on the smallest classes, tested for equivalence rather than for significance alone, and repeated on patch-disjoint splits.

Several limitations apply. The blocks were compared in two residual hosts of one architecture family, with one patch size and one feature pipeline; transformer hosts may use attention differently. No block was tuned individually: of two other learning rates, the higher one (15 seeds) reduced the Indian Pines gap to a non-significant 1.5 points at most and the lower one (five seeds) widened it, so a per-block tuning study could change the size of that single-scene effect. The deeper host is also wider, so it tests capacity rather than depth alone. On the patch-disjoint splits, 10–15 seeds can only exclude differences larger than about two points, and the cluster construction adds a spatial shift between training and test data, which the block-disjoint split reduces but does not remove; the standard IP split still lets patches overlap, and the standard PU and Houston splits were unavailable.

6. Conclusion

In lightweight hyperspectral CNNs, seven channel-attention blocks were equivalent within ± 1 point of average accuracy whenever every class had at least 15 training pixels, and on random splits of the large WHU-Hi scene. The one large difference, 2–4 points in favour of dispersion-based gates on random Indian Pines splits, came from two single-pixel classes. A deeper plain CNN (93.8% AA) matched the best shallow block (93.3%), the gap depended on the learning rate, it did not appear when the same scarcity was imposed on other scenes, and it was mostly not significant on patch-disjoint splits, where accuracy dropped by 32–45 points; on the milder block-disjoint split of Pavia University, four blocks, including both SNSM variants, were significantly worse than the plain CNN. There no block was consistently best, although differences

of about two points could not be excluded. Class counts, host network, training setting and evaluation protocol shaped reported accuracy more than the choice of attention block. **Before claiming a gain for a new block, report the training pixels of the smallest classes, test for equivalence rather than relying on non-significance, and repeat the comparison on patch-disjoint splits.**

Disclosure statement

The authors report there are no competing interests to declare.

Funding

No external funding was received for this research.

Data and code availability

The datasets are publicly available (Grupo de Inteligencia Computacional 2020); the standard IP split is distributed with SpectralFormer (Hong et al. 2022). Code, per-run results, the supplement and the scripts that generate every table and figure are available at `https://github.com/Devguru-codes/research_paper_code`.

Use of generative AI

Generative AI (Claude, Anthropic) was used to assist with writing experiment code, checking references and editing the manuscript text. The authors reviewed and verified all code, results and text, and take full responsibility for the content.

References

Ahmad, Muhammad, Manuel Mazzara, Salvatore Distefano, Adil Mehmood Khan, and Hamad Ahmed Altuwaijri. 2024. "Improving Generalization for Hyperspectral Image Classification: The Impact of Disjoint Sampling on Deep Models." *Computers, Materials & Continua* 81 (1): 503–532. https://doi.org/10.32604/cmc.2024.056318.

Alkhatib, Mohammed Q. 2026. "Data Leakage in Patch-Based Hyperspectral Image Classification: Quantifying the Impact of Spatial Overlap." arXiv preprint arXiv:2610.08770. https://doi.org/10.48550/arXiv.2610.08770.

Audebert, Nicolas, Bertrand Le Saux, and Sebastien Lefevre. 2019. "Deep Learning for Classification of Hyperspectral Data: A Comparative Review." *IEEE Geoscience and Remote Sensing Magazine* 7 (2): 159–173. https://doi.org/10.1109/MGRS.2019.2912563.

Baumgardner, Marion F., Larry L. Biehl, and David A. Landgrebe. 2015. "220 Band AVIRIS Hyperspectral Image Data Set: June 12, 1992 Indian Pine Test Site 3." Purdue University Research Repository. https://doi.org/10.4231/R7RX991C.

Bello, Irwan, William Fedus, Xianzhi Du, Ekin Dogus Cubuk, Aravind Srinivas, Tsung-Yi Lin, Jonathon Shlens, and Barret Zoph. 2021. "Revisiting ResNets: Improved Training and Scaling Strategies." In *Advances in Neural Information Processing Systems*, Vol. 34, 22614–22627.

Bioucas-Dias, Jose M., Antonio Plaza, Gustavo Camps-Valls, Paul Scheunders, Nasser Nasrabadi, and Jocelyn Chanussot. 2013. "Hyperspectral Remote Sensing Data Analysis and Future Challenges." *IEEE Geoscience and Remote Sensing Magazine* 1 (2): 6–36. https://doi.org/10.1109/MGRS.2013.2244672.

Bouthillier, Xavier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Nazanin Mohammadi Sepahvand, et al. 2021. "Accounting for Variance in Machine Learning Benchmarks." In *Proceedings of Machine Learning and Systems*, Vol. 3, 747–769.

Chen, Yushi, Hanlu Jiang, Chunyang Li, Xiuping Jia, and Pedram Ghamisi. 2016. "Deep Feature Extraction and Classification of Hyperspectral Images Based on Convolutional Neural Networks." *IEEE Transactions on Geoscience and Remote Sensing* 54 (10): 6232–6251. https://doi.org/10.1109/TGRS.2016.2584107.

Feng, Hao, Yongcheng Wang, Zheng Li, Ning Zhang, Yuxi Zhang, and Yunxiao Gao. 2023. "Information Leakage in Deep Learning-Based Hyperspectral Image Classification: A Survey." *Remote Sensing* 15 (15): 3793. https://doi.org/10.3390/rs15153793.

Gamba, Paolo. 2004. "ROSIS Pavia University Hyperspectral Data Set." Telecommunications and Remote Sensing Laboratory, University of Pavia. `https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes`.

Grupo de Inteligencia Computacional. 2020. "Hyperspectral Remote Sensing Scenes." University of the Basque Country (UPV/EHU). `https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes`.

Haut, Juan Mario, Mercedes E. Paoletti, Javier Plaza, Antonio Plaza, and Jun Li. 2019. "Visual Attention-Driven Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 57 (10): 8065–8080. https://doi.org/10.1109/TGRS.2019.2918080.

He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. "Deep Residual Learning for Image Recognition." In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. https://doi.org/10.1109/CVPR.2016.90.

Hong, Danfeng, Zhu Han, Jing Yao, Lianru Gao, Bing Zhang, Antonio Plaza, and Jocelyn Chanussot. 2022. "SpectralFormer: Rethinking Hyperspectral Image Classification With Transformers." *IEEE Transactions on Geoscience and Remote Sensing* 60: 1–15. https://doi.org/10.1109/TGRS.2021.3130716.

Hu, Jie, Li Shen, and Gang Sun. 2018. "Squeeze-and-Excitation Networks." In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 7132–7141. https://doi.org/10.1109/CVPR.2018.00745.

Hughes, G. 1968. "On the mean accuracy of statistical pattern recognizers." *IEEE Transactions on Information Theory* 14 (1): 55–63. https://doi.org/10.1109/TIT.1968.1054102.

Ioffe, Sergey, and Christian Szegedy. 2015. "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift." In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 448–456. https://doi.org/10.48550/arXiv.1502.03167.

Jolliffe, Ian T., and Jorge Cadima. 2016. "Principal component analysis: a review and recent developments." *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 374 (2065): 20150202. https://doi.org/10.1098/rsta.2015.0202.

Kakde, Deovrat, Arin Chaudhuri, Hansi Jiang, Carol Sadek, Seunghyun Kong, and Yuwei Liao. 2018. "A New Bandwidth Selection Criterion for Using SVDD to Analyze Hyperspectral Data." In *Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XXIV*, Proc. SPIE. https://doi.org/10.1117/12.2314964.

Lee, Hyunjae, Hyo-Eun Kim, and Hyeonseob Nam. 2019. "SRM: A Style-Based Recalibration Module for Convolutional Neural Networks." In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, 1854–1862. https://doi.org/10.1109/ICCV.2019.00194.

Li, Rui, Shunyi Zheng, Chenxi Duan, Yang Yang, and Xiqi Wang. 2020. "Classification of Hyperspectral Image Based on Double-Branch Dual-Attention Mechanism Network." *Remote Sensing* 12 (3): 582. https://doi.org/10.3390/rs12030582.

Li, Shutao, Weiwei Song, Leyuan Fang, Yushi Chen, Pedram Ghamisi, and Jon Atli

Benediktsson. 2019. "Deep Learning for Hyperspectral Image Classification: An Overview." *IEEE Transactions on Geoscience and Remote Sensing* 57 (9): 6690–6709. https://doi.org/10.1109/TGRS.2019.2907932.

Liang, Jie, Jun Zhou, Yuntao Qian, Lian Wen, Xiao Bai, and Yongsheng Gao. 2017. "On the Sampling Strategy for Evaluation of Spectral-Spatial Methods in Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 55 (2): 862–880. https://doi.org/10.1109/TGRS.2016.2616489.

Loshchilov, Ilya, and Frank Hutter. 2019. "Decoupled Weight Decay Regularization." In *International Conference on Learning Representations (ICLR)*, https://doi.org/10.48550/arXiv.1711.05101.

Makantasis, Konstantinos, Konstantinos Karantzalos, Anastasios Doulamis, and Nikolaos Doulamis. 2015. "Deep supervised learning for hyperspectral data classification through convolutional neural networks." In *2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 4959–4962. https://doi.org/10.1109/IGARSS.2015.7326945.

Mallat, S.G. 1989. "A theory for multiresolution signal decomposition: the wavelet representation." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11 (7): 674–693. https://doi.org/10.1109/34.192463.

Mei, Xiaoguang, Erting Pan, Yong Ma, Xiaobing Dai, Jun Huang, Fan Fan, Qinglei Du, Hong Zheng, and Jiayi Ma. 2019. "Spectral-Spatial Attention Networks for Hyperspectral Image Classification." *Remote Sensing* 11 (8): 963. https://doi.org/10.3390/rs11080963.

Musgrave, Kevin, Serge Belongie, and Ser-Nam Lim. 2020. "A Metric Learning Reality Check." In *Computer Vision – ECCV 2020*, Vol. 12370 of *Lecture Notes in Computer Science*, 681–699. https://doi.org/10.1007/978-3-030-58595-$2_4$1.

Nalepa, Jakub, Michal Myller, and Michal Kawulok. 2019. "Validating Hyperspectral Image Segmentation." *IEEE Geoscience and Remote Sensing Letters* 16 (8): 1264–1268. https://doi.org/10.1109/LGRS.2019.2895697.

Paszke, Adam, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, et al. 2019. "PyTorch: An Imperative Style, High-Performance Deep Learning Library." In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 8024–8035. https://doi.org/10.48550/arXiv.1912.01703.

Roy, Swalpa Kumar, Gopal Krishna, Shiv Ram Dubey, and Bidyut B. Chaudhuri. 2020. "HybridSN: Exploring 3-D–2-D CNN Feature Hierarchy for Hyperspectral Image Classification." *IEEE Geoscience and Remote Sensing Letters* 17 (2): 277–281. https://doi.org/10.1109/LGRS.2019.2918719.

Roy, Swalpa Kumar, Suvojit Manna, Tiecheng Song, and Lorenzo Bruzzone. 2021. "Attention-Based Adaptive Spectral–Spatial Kernel ResNet for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 59 (9): 7831–7843. https://doi.org/10.1109/TGRS.2020.3043267.

Srivastava, Nitish, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014. "Dropout: A Simple Way to Prevent Neural Networks from Overfitting." *Journal of Machine Learning Research* 15 (56): 1929–1958. `https://jmlr.org/papers/v15/srivastava14a.html`.

Sun, Le, Guangrui Zhao, Yuhui Zheng, and Zebin Wu. 2022. "Spectral–Spatial Feature Tokenization Transformer for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 60: 1–14. https://doi.org/10.1109/TGRS.2022.3144158.

Wang, Qilong, Banggu Wu, Pengfei Zhu, Peihua Li, Wangmeng Zuo, and Qinghua Hu. 2020. "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks." In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 11531–11539. https://doi.org/10.1109/CVPR42600.2020.01155.

Woo, Sanghyun, Jongchan Park, Joon-Young Lee, and In So Kweon. 2018. "CBAM: Convolutional Block Attention Module." In *Computer Vision – ECCV 2018, Lecture Notes in Computer Science, vol. 11211*, 3–19. https://doi.org/10.1007/978-3-030-01234-2_1.

Yang, Zongxin, Linchao Zhu, Yu Wu, and Yi Yang. 2020. "Gated Channel Transformation for Visual Recognition." In *2020 IEEE/CVF Conference on Computer Vision and Pattern*

Recognition (CVPR), 11791–11800. https://doi.org/10.1109/CVPR42600.2020.01181.

Zhong, Yanfei, Xin Hu, Chang Luo, Xinyu Wang, Ji Zhao, and Liangpei Zhang. 2020. "WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF." *Remote Sensing of Environment* 250: 112012. https://doi.org/10.1016/j.rse.2020.112012.

Zhong, Zilong, Jonathan Li, Zhiming Luo, and Michael Chapman. 2018. "Spectral–Spatial Residual Network for Hyperspectral Image Classification: A 3-D Deep Learning Framework." *IEEE Transactions on Geoscience and Remote Sensing* 56 (2): 847–858. https://doi.org/10.1109/TGRS.2017.2755542.

Zhu, Minghao, Licheng Jiao, Fang Liu, Shuyuan Yang, and Jianing Wang. 2021. "Residual Spectral–Spatial Attention Network for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 59 (1): 449–462. https://doi.org/10.1109/TGRS.2020.2994057.